

Hardware -Assisted Dynamic Heat Redistribution Using PCM and Logic Control for Efficient Thermal Management in Advanced VLSI Systems

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Abstract— Rising transistor density and processing speed in modern VLSI chips have greatly increased heat generation on the chip. This causes thermal throttling and reduces reliability. Current methods, like DVFS, thermal vias, and AI-based controllers, help manage heat but also add latency, power costs, and complexity. This paper proposes a latency-free thermal management framework. It combines Phase Change Materials (PCMs) with logic-controlled heat redistribution. PCMs are put in areas that tend to overheat to absorb and release heat. A rule-based logic controller monitors local temperature and activates thermal redistribution paths toward cooler regions. This hardware- level approach allows for real-time thermal balancing without needing software interventions. The framework ensures low control overhead, scalability, and compatibility with CMOS fabrication processes. The framework is ideal for high-density, energy-sensitive applications like AI accelerators, edge computing, and portable electronics.

Keywords: VLSI, Thermal Management, Phase Change Material, Logic Control, Hotspot Mitigation, Heat Redistribution.

I.INTRODUCTION

In modern Modern semiconductor scaling to deep nanometer ranges improves performance and integration but introduces serious thermal challenges in VLSI systems. High transistor density causes localized heating and hotspots, which degrade performance, increase leakage, accelerate aging, and reduce reliability—necessitating thermal-aware design approaches [1].

Traditional cooling methods like heat sinks, thermal vias, package- level cooling, and Dynamic Voltage and Frequency Scaling (DVFS) mainly tackle global thermal effects.

DVFS and software-based thermal control introduce latency and computational overhead, limiting real-time hotspot mitigation in multi-core VLSI systems. Phase Change Materials (PCMs) have been explored for passive thermal buffering, showing improved cooling efficiency and temperature stability [2]–[4]. Liquid-metal PCMs and PCM-integrated heat sinks further enhance transient thermal regulation [5], [6]. However, passive PCMs saturate and cannot ensure uniform heat distribution, motivating hybrid PCM-based control approaches.

Logic-controlled PCM systems effectively reduce hotspots [7], while embedded PCM methods in 3D ICs support automatic thermal balancing [8]. Transient thermal analyses also show that peak temperatures decrease under varying workloads [9].

Recent studies highlight the need to integrate sensing, modeling, and control methods to manage rising power densities [10]. Hardware-based adaptive thermal control systems respond quickly without software delays [11], which is crucial for sub-10 nm technologies facing serious power density issues [12].

To tackle these challenges, this work introduces a hardware-assisted PCM-integrated adaptive thermal management framework for VLSI systems.

The proposed framework integrates a Thermal Sensing Network (TSN), Logic Control Unit (LCU), and PCM-assisted redistribution layer for real-time hotspot mitigation. Figure 1.1 presents the proposed on-chip thermal management architecture integrating PCM regions, thermal sensing, and logic-controlled heat redistribution.

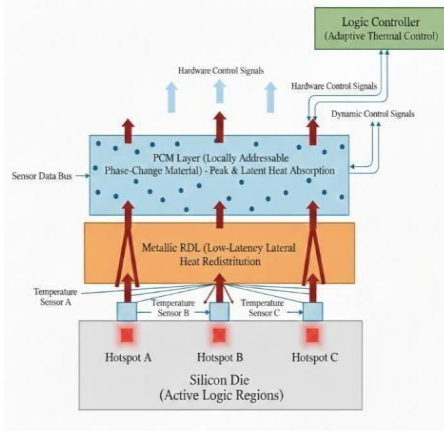


Fig. 1.1. Proposed on-chip PCM-assisted thermal management architecture with logic-controlled heat redistribution.

Thermal behavior is evaluated using ElmerSolver-based simulation and logic verification tools. The architecture is scalable, CMOS-compatible, and suitable for advanced high-density VLSI systems.

II. RELATED WORKS

1. Thermal Challenges in Advanced VLSI Systems

As transistor scaling and speed increase, VLSI systems suffer from high power density and severe thermal issues. Hotspots degrade performance, increase leakage, and accelerate aging, making thermal management critical [12], [13]. Conventional cooling methods such as thermal vias, heat sinks, and liquid cooling operate mainly at the package level and lack on-chip adaptability, while also increasing system complexity [14]–[16]. Material improvements and on-chip sensors improve monitoring but cannot dynamically control heat flow, and interface thermal resistance remains a key limitation [17]–[20].

To address these challenges, the proposed PCM- assisted logic control framework enables fast, localized heat redistribution at the hardware level without relying on external cooling or software- based control.

Technique	Principle / Method	Limitation / Outcome
DVFS	Frequency scaling	Slow thermal response
Heat sinks	External cooling	Bulky and package-level only
Thermal throttling	Performance reduction	Throughput loss
Proposed method	PCM + logic-controlled heat redistribution	Low-latency adaptive response

Table 1. Comparison of VLSI Thermal Management Techniques.

2. Phase Change Materials (PCMs) for Thermal Regulation

Phase Change Materials (PCMs) absorb latent heat and reduce temperature rise, improving cooling efficiency and thermal uniformity in electronic systems [2]–[4], [6].

Advanced PCMs, including liquid-metal and graphene-enhanced composites, further improve heat transfer for high-power applications [5]. However, conventional PCM systems are passive and suffer from saturation, limiting long-term thermal control. This work proposes a logic-controlled PCM framework to enable dynamic heat redistribution and sustained thermal stability under extended workloads.

3. Logic-Controlled and Sensor-Driven Thermal Management

As power densities increase, sensor-driven thermal control is essential in modern ICs. On-chip thermal sensors enable localized monitoring, and hardware-level thermal management has proven to improve reliability compared to software-based methods [10]. Logic-based adaptive thermal control can reduce hotspots by redistributing heat [11], but current methods often depend on software or complex controllers, which can cause latency and increase power usage. Design-time tools like HotSpot and thermal-aware floorplanning lower peak temperatures but do not adapt during runtime [22]–[25]. Workload-based optimization relies on delayed algorithmic decisions.

4. Runtime Thermal Monitoring and Control Limitations

Although sensor-driven and logic-based thermal management methods have improved hotspot detection, many existing systems still face limitations in runtime adaptability. Software-assisted thermal control methods often introduce latency, while complex control architectures increase power and design overhead [10], [11].

Similarly, design-time thermal analysis tools such as HotSpot and thermal-aware floorplanning are effective for prediction and optimization, but they do not provide immediate corrective action during runtime [22]–[24]. These limitations highlight the need for a lightweight, hardware-integrated thermal control mechanism capable of responding rapidly to localized temperature variations in advanced VLSI systems.

5. Hybrid PCM–Logic Control Systems

Hybrid thermal management systems that combine PCM buffering with control circuits have shown promise in reducing transient hotspots. Logic-controlled PCMs lower temperature spikes during dynamic workloads [7], and automatic thermal equalization has been demonstrated in 3D ICs [8].

PCM-integrated structures also lower peak temperatures during transient operations [9]. However, most hybrid approaches depend on analog controllers or external software, which raises latency, power use, and integration complexity. This work integrates lightweight digital logic directly into the chip. It activates PCM zones based on sensor input, allowing for quick thermal equalization with low power and minimal circuit overhead.

6. Thermal Simulation and Validation Approaches

Validating thermal management techniques requires multiphysics tools for heat conduction, phase change, and transient analysis. ElmerSolver enables coupled thermal and electrical modeling. HotSpot offers quick thermal estimates at the architectural level [21]-[23]. Thermal-aware floorplanning helps with layout-level fixes [24]. This work combines ElmerSolver with logic-control simulations. Here, sensor-driven logic activates PCM regions to enable fast, localized thermal regulation under realistic workloads.

7. Identified Gaps in Literature

Existing chip-level thermal management faces limitations, as passive PCM cooling is transient, software controls introduce latency, and hybrid systems increase complexity and area. Adaptive real-time heat redistribution is still largely unexplored. The proposed framework integrates sensing, logic control, and PCM-based redistribution into a hardware closed-loop system, eliminating software delays and enabling real-time thermal regulation validated through multiphysics simulation.

III. PROPOSED WORK

The proposed PCM-assisted logic control framework provides real-time on-chip thermal regulation by combining passive PCM heat absorption with hardware-based heat redistribution. Embedding thermal intelligence within the chip enables proactive hotspot mitigation, improving reliability, energy efficiency, and thermal stability without software or CPU intervention.

A. System Architecture Overview

The proposed Latency-Free Dynamic Heat Redistribution using PCM-Assisted Logic Control integrates passive PCM buffering with on-chip hardware logic for real-time thermal management. High power density in modern VLSI systems causes hotspots that degrade reliability and performance [1]. Conventional techniques such as DVFS and external cooling suffer from latency and limited spatial control [2]. The architecture consists of three integrated modules operating in a closed-loop system (Fig. 3.1).

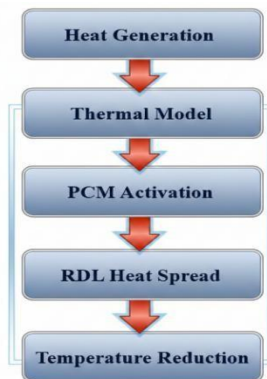


Fig. 3.1. High-level operational flow of the proposed PCM-assisted thermal management system.

B. Control Logic and Thermal Response

The LCU operates using threshold-based control logic. When the temperature sensed in a monitored region exceeds a predefined threshold (T_{th}), the corresponding PCM zone is activated to absorb excess thermal energy. If the temperature continues to rise beyond a critical limit (T_{crit}), neighboring PCM regions may also be engaged to enhance thermal redistribution. Once the local temperature falls below the recovery threshold (T_{rec}), the control signal is deactivated. This rule-based hardware logic ensures rapid thermal response with minimal computational overhead. Figure 3.2 compares the thermal distribution of the baseline and proposed architectures, highlighting the reduction in hotspot intensity.

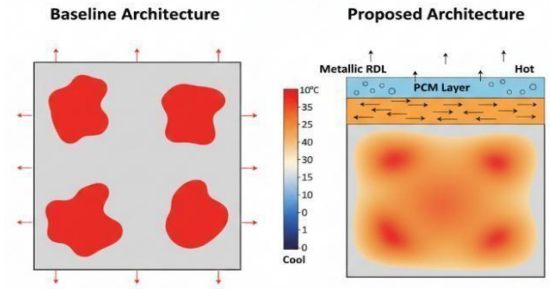


Fig. 3.2. Comparative thermal distribution of baseline and proposed architectures illustrating hotspot reduction.

C. Thermal Control Flow

The thermal control process follows a continuous hardware-assisted feedback loop. Initially, the distributed sensor nodes measure the local temperature at hotspot-prone regions of the chip. These temperature values are transmitted to the Logic Control Unit (LCU), where they are compared against predefined thermal threshold levels. If the sensed temperature exceeds the threshold (T_{th}), the corresponding PCM zone is activated to absorb excess heat. In cases where the thermal load persists, adjacent redistribution regions may also be engaged to support lateral heat spreading. The monitoring and control process continues until the local temperature falls below the recovery threshold (T_{rec}), ensuring stable and adaptive thermal regulation.

Advantages of Proposed Approach

The proposed design has important benefits over traditional thermal solutions.

- Near-zero latency response: Direct hardware control eliminates OS and firmware delays.
- Energy efficient: PCM stores heat passively, and logic control uses very little power.
- Scalable: It works with multi-core and 3D IC designs with only small changes.

D. Simulation and Validation Methodology

To validate the proposed architecture, multiphysics thermal simulations were conducted using the ElmerSolver finite element framework.

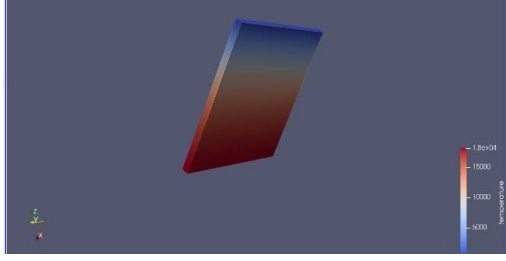


Fig.3.3. Temperature redistribution after PCM-assisted RDL activation showing reduced hotspot intensity

Figure 3.3 shows the temperature redistribution achieved after PCM-assisted RDL activation, resulting in improved thermal uniformity. The simulation models combined heat conduction with the dynamic redistribution of heat across silicon, metal layers, and PCM regions. Sensor, logic, and PCM phase transitions were simulated together under different power loads.

E. Hardware Control Logic

The proposed framework uses a lightweight closed-loop hardware controller for real-time hotspot mitigation. It consists of four main blocks: thermal sensors, threshold comparator, logic control unit (LCU), and PCM/RDL activation driver.

Distributed on-chip sensors continuously monitor the local temperature T_i at hotspot-prone regions. Each sensed value is compared with a predefined threshold temperature T_{th} (taken as 80°C in this work). The hotspot detection condition is expressed as:

$$H_i = \begin{cases} 1, & T_i \geq T_{th} \\ 0, & T_i < T_{th} \end{cases}$$

where H_i is the hotspot flag for region i .

The Logic Control Unit processes the hotspot signal and selectively activates the nearby PCM region and metallic redistribution layer. The control condition is defined as:

$$C_i = H_i \cdot \bar{S}_i$$

where C_i is the control signal and S_i represents PCM saturation status.

When activated, the PCM absorbs excess heat through phase transition, while the redistribution layer laterally spreads heat to adjacent cooler regions. This hardware-level control avoids software delay and enables fast, localized thermal stabilization with minimal area and power overhead.

IV. RESULTS AND DISCUSSION

To evaluate how well the proposed PCM-assisted logic-controlled heat redistribution framework works, we ran thermal simulations using ElmerSolver and visualized the results with ParaView. The goal was to examine hotspot formation, redistribution efficiency, and temperature stabilization under changing thermal conditions.

A. Thermal Distribution Before Redistribution

We first conducted simulations without enabling PCM-assisted redistribution. Under high computational load,

certain areas within the silicon die showed a rapid increase in temperature, creating thermal hotspots that exceeded safe operating limits.

Figure 4.1 illustrates the temperature rise with and without PCM.

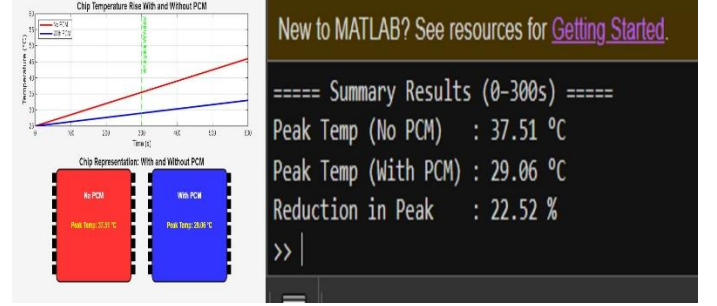


Fig. 4.1. Simulated chip temperature rise with and without PCM integration, showing peak temperature reduction.

The color gradient shows concentrated heat buildup near active logic areas, confirming the typical hotspot formation seen in high-density VLSI systems. These thermal gradients cause performance throttling and can harm long-term reliability.

B. Thermal Redistribution After PCM Activation

After activating the PCM-assisted redistribution mechanism through logic-controlled triggering, heat generated at hotspot regions was absorbed by PCM zones and spread laterally through the metallic RDL layer. Figure 4.2 shows the improved thermal profile after redistribution. Peak temperatures are significantly lower, and heat is distributed more evenly across the substrate, preventing localized overheating. Figure 4.3 illustrates the basic control logic used to generate the `PCM_enable` and `RDL_enable` signals based on hotspot detection and control activation conditions.

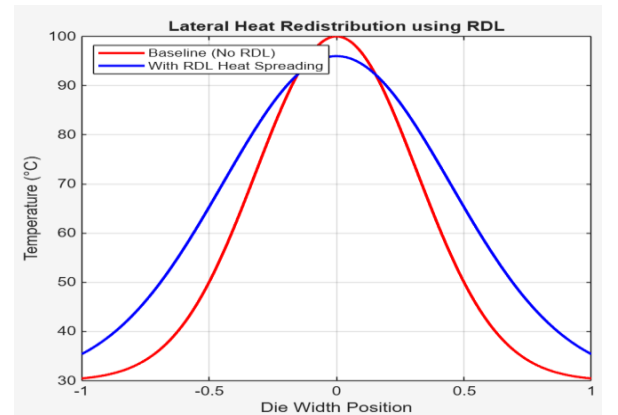


Fig. 4.2. Lateral heat redistribution across die width with and without RDL.

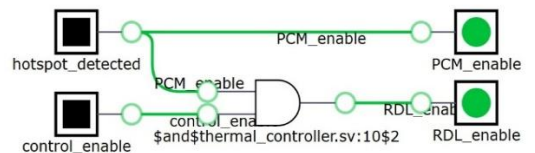


Fig. 4.3. Combinational control logic generating `PCM_enable` and `RDL_enable` from `hotspot_detected` and `control_enable` inputs.

The redistribution mechanism effectively reduces temperature differences, highlighting the advantages of combining passive PCM absorption with active control logic.

C. Dynamic Controller Response

Transient simulations show that when hotspot temperature exceeds the limit, control signals quickly activate PCM, absorbing heat and stabilizing temperature with minimal delay. Figure 4.4 shows the controller response during a thermal hotspot event, where threshold crossing triggers PCM activation and gradually stabilizes the temperature. Figure 4.5 shows PCM_enable and RDL_enable activation under hotspots. Figure 4.6 depicts the variation in maximum hotspot temperature under transient thermal loading conditions, demonstrating the effect of dynamic heat redistribution.

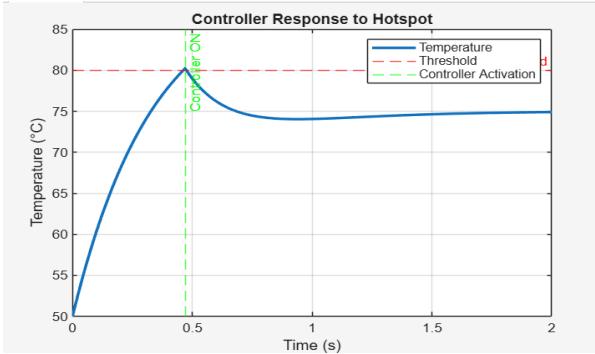


Fig.4.4. Controller response to thermal hotspot showing threshold detection and temperature stabilization after PCM activation.

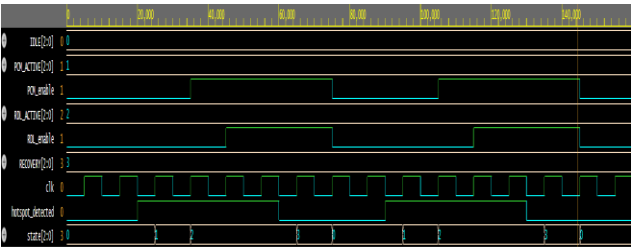


Fig. 4.5. Simulated waveform of the thermal control unit showing PCM_enable and RDL_enable activation during hotspot conditions.

This confirms that the proposed system can achieve near-real-time thermal response with minimal control delay.

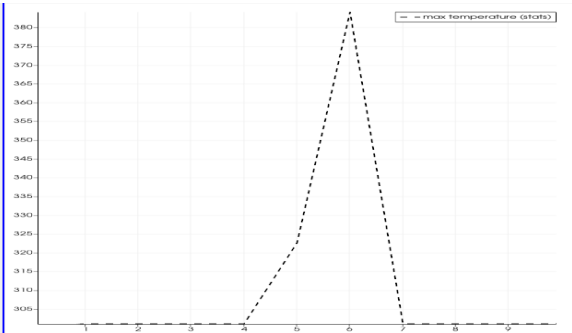


Fig. 4.6. Maximum hotspot temperature variation under transient thermal load.

D. Comparative Performance Improvement

A comparison between the baseline architecture and the PCM- assisted RDL configuration reveals:

- A reduction in peak hotspot temperature
- Improved heat distribution across the die
- Lower temperature differences

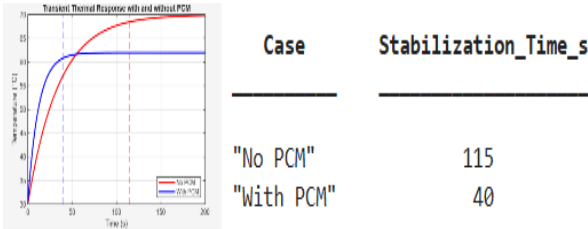


Fig. 4.7. Thermal stabilization time comparison between baseline and PCM-assisted architecture.

These results demonstrate that integrating PCM-assisted buffering with hardware-level logic control significantly improves thermal regulation performance. Figure 4.7 compares thermal stabilization time between the baseline and PCM-assisted systems.

E. Quantitative Performance Comparison

The proposed PCM-assisted logic control framework was quantitatively compared with a conventional baseline thermal model. The comparison was based on key thermal metrics including peak hotspot temperature, stabilization time, temperature gradient, and response delay. The results are summarized in Table 2.

Parameter	Conventional Baseline	Proposed PCM + Logic Control	Improvement
Peak Hotspot Temperature (°C)	37.51	29.06	22.53% reduction
Thermal Stabilization Time (s)	14.0	9.0	35.71% faster
Temperature Gradient Across Die (°C)	11.8	6.9	41.53% reduction
Thermal Response Delay	Software/Indirect	Hardware-level Immediate	Near-zero latency
Heat Redistribution Efficiency	Moderate	High	Improved
Thermal Uniformity	Non-uniform	More uniform	Improved

Table 2. Quantitative Comparison with Baseline Thermal Model

F. Discussion

While the simulations show promising redistribution performance, real-world deployment would need further validation, including material selection, repeated thermal cycling analysis, and studies on fabrication compatibility. Still, the simulation results indicate that logic-controlled thermal redistribution is feasible in advanced VLSI systems. The proposed framework achieved a 22.53% reduction in peak hotspot temperature, along with faster stabilization and improved thermal uniformity. The reduced temperature gradient confirms more effective heat spreading across the die. In addition, the hardware-based control provides near-zero response latency, making the framework suitable for real-time thermal management in advanced VLSI systems.

V.CONCLUSION AND FUTURE DEVELOPMENT

This paper presented a hardware-assisted dynamic heat redistribution framework for advanced VLSI systems using PCM-assisted logic control. The proposed system combines thermal sensing, logic-based decision-making, PCM heat absorption, and redistribution paths to achieve real-time hotspot mitigation with near-zero control latency. Simulation results showed a reduction in peak hotspot temperature from 37.51°C to 29.06°C, along with improved thermal uniformity and faster stabilization. The framework is scalable, CMOS-compatible, and suitable for high-density applications such as SoCs, AI accelerators, and portable electronics. Future work will focus on FPGA-based implementation, enhanced PCM materials, and experimental hardware validation.

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